WASP-35b

R-1665

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-35_151202` · created 2026-08-02T10:03:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457358.8669 +/- 0.00055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.064 +/- 0.044
Transit depth (Rp/Rs)^2	0.41 +/- 0.56 [%]
Orbital Inclination (inc)	83.1 +/- 2.06
Airmass coefficient 1 (a1)	1.23 +/- 0.19
Airmass coefficient 2 (a2)	-0.125 +/- 0.052
Scatter in the residuals of the lightcurve fit is	15.24 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.048 +/- 0.044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.064 ± 0.044 vs 0.1241 (deviation 1.27σ)
Duration fit vs published	0.048 d vs 0.1317 d (deviation 1.77σ)
Mid-time O–C	0.7 min
Depth SNR	1.4
Residual scatter	15.24 %

Light curve

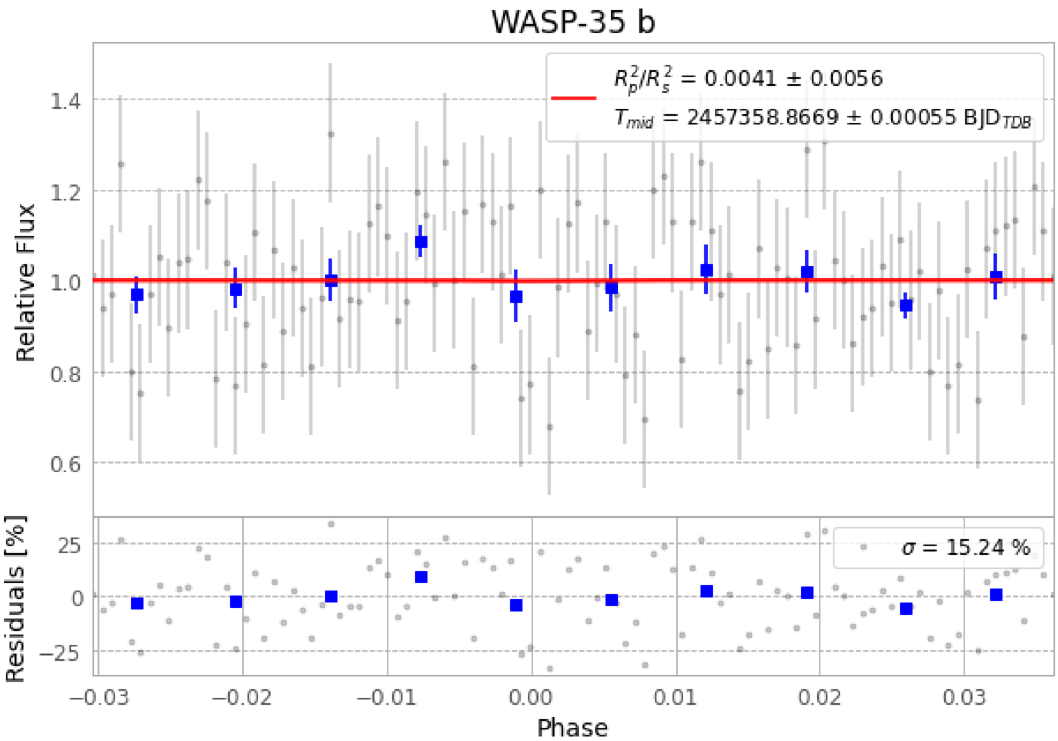

Figure 1 — FinalLightCurve_WASP-35 b_2015-12-01.png (SHA-256 0e00ceb0899f2760…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [33, 127], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 269d25b78c421cac…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

102 FITS frames (67 MB), `WASP-35151202062112.FITS` → `WASP-35151202112412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-151202.log (dcb85a75ae5c…), FinalLightCurve_WASP-35 b_2015-12-01.png (0e00ceb0899f…), FinalLightCurve_WASP-35 b_2015-12-01.csv (3f8e04affc9e…)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)